

MISCELLANY :→

① There are two married couples that need to cross a river. A small boat is available that will hold just two people at a time. The males involved are quite jealous. No woman can be left with a man unless her husband is also present. There are no other constraints. How can these four people cross the river? What is the fewest number of trips possible?

⇒ Let us denote the people by H_1, H_2 for the husbands and W_1, W_2 for the corresponding wives.

⇒ H_1 and W_1 cross the river on the first trip.

⇒ H_1 returns.

⇒ H_1 and H_2 cross the river.

⇒ H_2 returns.

⇒ H_2 picks up his wife W_2 and then crosses the river.

② A general needs to take his troops across the river. He spies two boys with a small boat. He commands both the boat and the boys. Unfortunately, the boat will only hold two boys or one soldier. Yet he determines a method for getting his troops across. What could it be?

⇒ Notice that the number of soldiers is not given, suggesting that the problem is independent of the number of soldiers.

⇒ For the first trip, there is no sense to send one soldier. For all he could do would be to go across and then either

(i) stay and leave everyone else stranded or (ii) just row back.

⇒ For the first trip, the two boys go. One boy returns. Then one soldier goes across. Then the boy on the far side of the river can take the boat back.

⇒ Now matters are just as at the start, except that one soldier has been transferred across the river.

⇒ So, now the two boys cross again. One boy stays on the far shore (with the lone soldier) and the other returns with the boat. Now a second soldier can go across. The boy returns with the boat. Now both boys are on the starting shore with the remaining soldiers.

⇒ Clearly, this process can be repeated indefinitely to get all the soldiers, and the commander, across.

③ A railroad engine can just couple with, then either pull or push, either one or two cars. Of course the two cars can be coupled to each other as well.

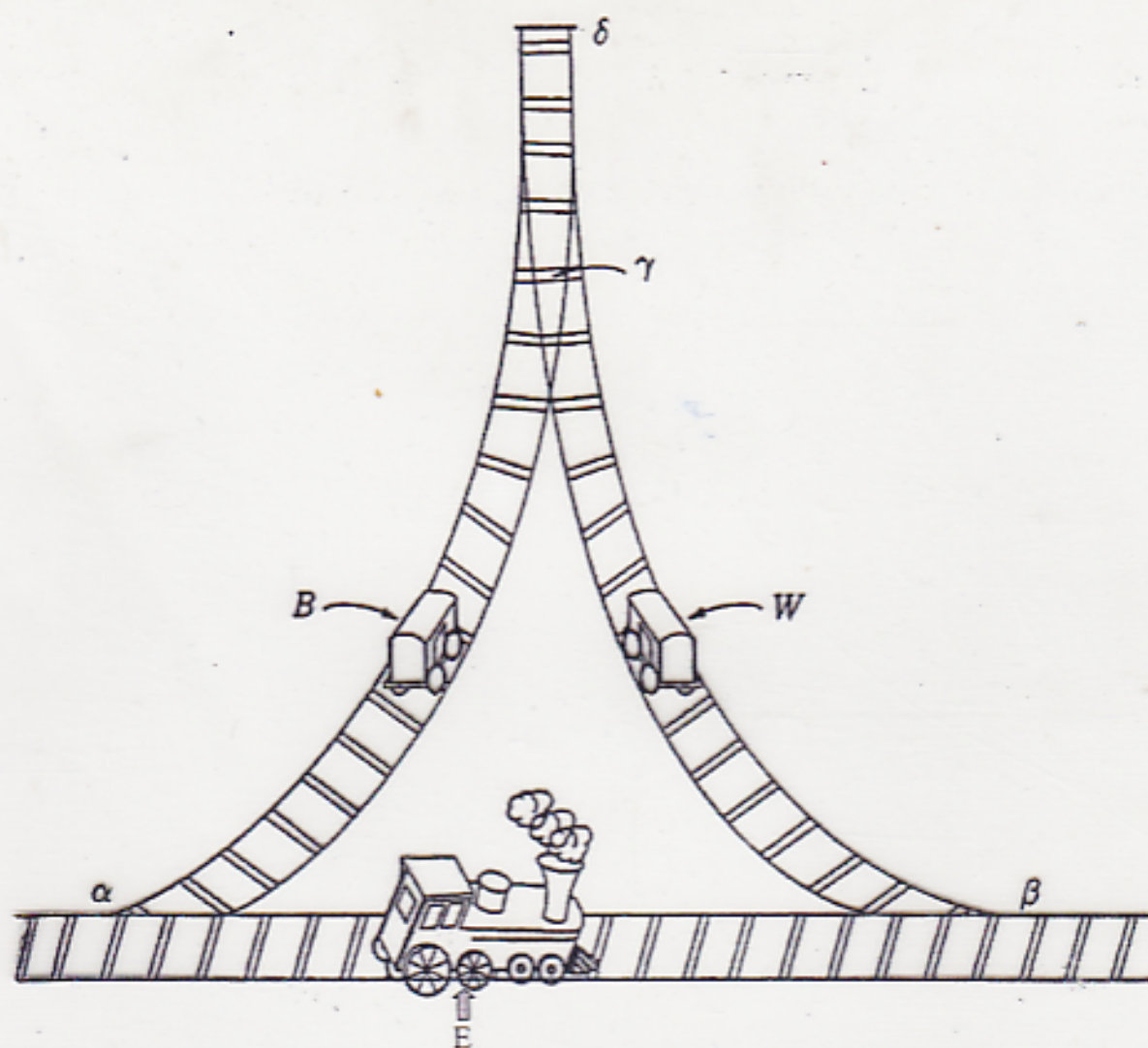
At a certain junction, the railroad track is configured as in Figure next slide. Note that the portion of track

between r and s in figure is a dead end and can hold either just one car or just the engine. However, the portion of the track to the left of X and the portion of the track to the right of β have no restrictions and can hold any number of cars.

How can the engine reverse the positions of the black car and the white car (that is, put the black car on the right track b/w β and r and the white car on the left track b/w X and r) and return to its original position facing right? This should be done in at most 10 moves. Here, a move consists either of having the engine go to some point and couple with a car, or having the engine pull a car to some point and uncouple.

⇒ Call the white car W , the black car B , and the engine E .

- (i) E drives past β , backs into βs , and couples with the white car.
- (ii) E backs the white car into $r s$, uncouples, and moves out along βs .
- (iii) E moves past β , backs along $X\beta$, goes forward into Xr , and couples its front end on the black car.
- (iv) E pushes B forward, couples it on W , and backs out past X .



- (v) E pushes both cars forward until W is midway between α and β , then uncouples ~~W~~ W.
- (vi) E backs past α , then pushes B into $\alpha\gamma$ and up into $\gamma\delta$, where it uncouples B.
- (vii) E backs down $\alpha\gamma$ past α , then moves forward on $\alpha\beta$ and couples with W.
- (viii) E backs on $\alpha\beta$ past α , then goes forward on $\alpha\gamma$, pushing W to the center of $\alpha\gamma$, where it uncouples W.
- (ix) E backs down $\alpha\gamma$, past α , moves forward on $\alpha\beta$ past β , backs into $\beta\gamma$ all the way to B, and couples with B.
- (x) E pulls B forward, decouples in the middle of $\beta\gamma$, continues forward past β , and then backs down $\alpha\beta$ to the midpoint of $\alpha\beta$.

ASSIGNMENT SOLUTION

① Calculate the sum of the first K odd integers.

We know,

$$1+2+3+\dots+K = \frac{K^2+K}{2}$$

We want to find:

$$1+3+5+7+\dots+(2K-1)$$

$$\text{The first integer } 1 = 2 \cdot 1 - 1$$

$$\text{The second integer } 3 = 2 \cdot 2 - 1$$

The third integer $5 = 2 \cdot 3 - 1$

The k th integer $= 2 \cdot k - 1$

⇒ Hence the last term is $2k-1$ in the above equation.

⇒ Adding all numbers from 1 to $(2k-1)$ and then subtracting the even numbers in this sum gives the sum of the odd numbers from 1 to $(2k-1)$

$$[1+2+3+4+\dots+(2k-1)] - [2+4+6+\dots+(2k-2)]$$

$$\Rightarrow [1+2+3+4+\dots+(2k-1)] - 2[1+2+3+\dots+(k-1)]$$

$$\Rightarrow \left[\frac{(2k-1)^2 + 2k-1}{2} \right] - 2 \left[\frac{(k-1)^2 + (k-1)}{2} \right]$$

$$\Rightarrow \left[\frac{(2k-1)(2k-1+1)}{2} \right] - 2 \left[\frac{(k-1)(k-1+1)}{2} \right]$$

$$\Rightarrow \frac{2k(2k-1)}{2} - \frac{2k(k-1)}{2}$$

$$\Rightarrow \frac{2k}{2} [2k-1-k+1]$$

$$\Rightarrow k[k]$$

$$\Rightarrow k^2$$

Hence the required sum is k^2 .

② How many zeroes end the number

$$\begin{aligned} 2^{300} \cdot 5^{600} \cdot 4^{400} &= 2^{300} \cdot 5^{600} \cdot (2^2)^{400} \\ &= 2^{300} \cdot 5^{600} \cdot 2^{800} = 2^{1100} \cdot 5^{600} \\ &= (2 \cdot 5)^{600} \cdot 2^{500} \\ &= 10^{600} \cdot 2^{500} \end{aligned}$$

(ends with 600 zeroes)

③ How many digits are used to number the pages of a book having 100 pages - numbered from 1 to 100?

Between 1 and 100, we have

$$\begin{array}{rcl} 9 \text{ numbers with 1 digit} & = & 9 \text{ digits} \\ 90 \text{ numbers with 2 digits} & = & 180 \text{ digits} \\ 1 \text{ number with 3 digits} & = & 3 \text{ digits} \\ \hline & & 192 \text{ digits} \end{array}$$

④ A watermelon weighs 500 pounds. It turns out that 99% of the weight of the watermelon is due to water. After the watermelon has sat in a drying room for a while, it turns out that it is only 98% water by weight. How much does it weigh now?

$$99\% \text{ of } 500 \text{ pounds} = 495 \text{ pounds of water}$$

⇒ So, we have 5 pounds that are not water.

⇒ After the watermelon has sat in a drying room for a while, this 5 pounds constitute 2%

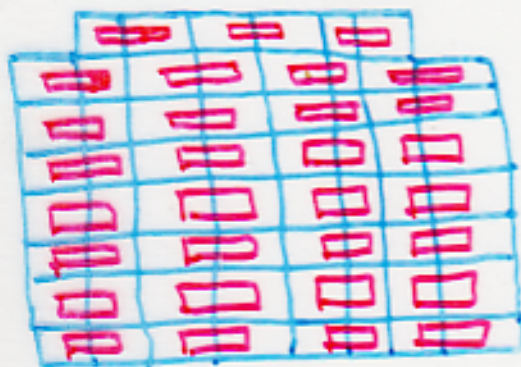
$$2\% \text{ is } 5 \text{ pounds}$$

$$1\% \text{ is } \frac{5}{2} \text{ pounds}$$

$$100\% \text{ is } \frac{5}{2} \times 100 = 250 \text{ pounds}$$

Hence it weighs 250 pounds now.

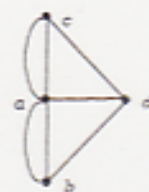
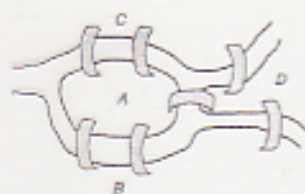
⑤ Examine the example from the text about tiling the bathroom floor. What happens if the two omitted squares are in adjacent corners of the bathroom instead of opposite corners?



Possible

Euler Paths and Circuits

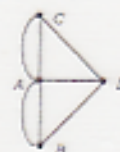
- The Seven bridges of Königsberg



Euler Paths and Circuits

- An *Euler path* is a path using every edge of the graph G exactly once.
- An *Euler circuit* is an Euler path that returns to its start.

Does this graph have an Euler circuit?
No.



Necessary and Sufficient Conditions

- How about multigraphs?
- A connected multigraph has a Euler circuit iff *each of its vertices has an even degree*.
- A connected multigraph has a Euler path but not an Euler circuit iff *it has exactly two vertices of odd degree*.

Example

- Which of the following graphs has an Euler circuit?



yes
(a, c, c, d, e, b, a)



no



no

Example

- Which of the following graphs has an Euler path?



yes
(a, c, c, d, e, b, a)



no



yes
(a, c, d, e, b, d, a, b)

Euler Circuit in Directed Graphs

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H_1

NO



H_2

(a, g, c, b, g, e, d, f, a)

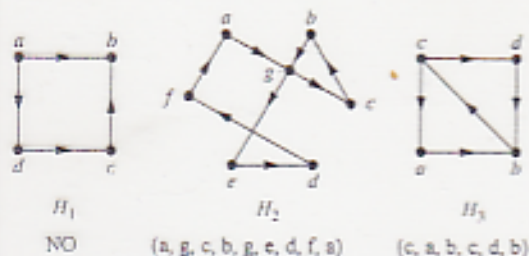


H_3

NO

Euler Path in Directed Graphs

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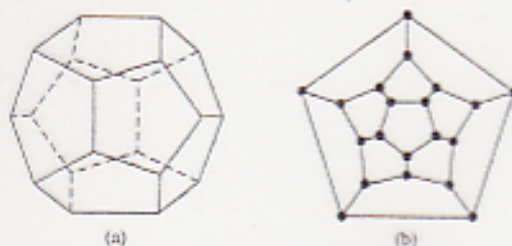


Hamilton Paths and Circuits

- A *Hamilton path* in a graph G is a path which visits every vertex in G exactly once.
- A *Hamilton circuit* is a Hamilton path that returns to its start.

Hamilton Circuits

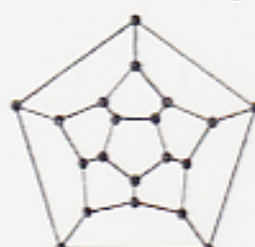
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Is there a circuit in this graph that passes through each vertex exactly once?

Hamilton Circuits

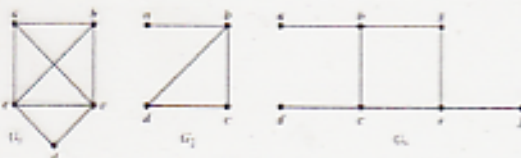
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Yes; this is a circuit that passes through each vertex exactly once.

Finding Hamilton Circuits

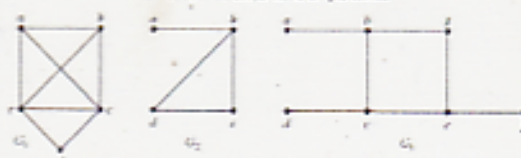
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Which of these three figures has a Hamilton circuit?
Of, if no Hamilton circuit, a Hamilton path?

Finding Hamilton Circuits

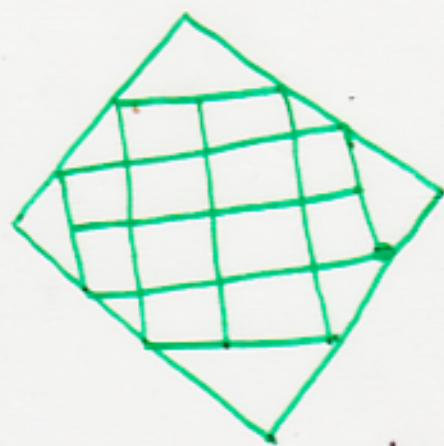
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- G_1 has a Hamilton circuit: a, b, c, d, e, a
- G_2 does not have a Hamilton circuit, but does have a Hamilton path: a, b, c, d
- G_3 has neither.

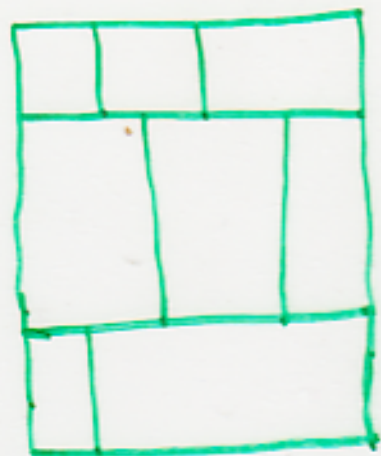
Impossible Problems :->

Explain why the line drawing in the figure below cannot be traced with a single, continuous, non-overlapping stroke of the pen.



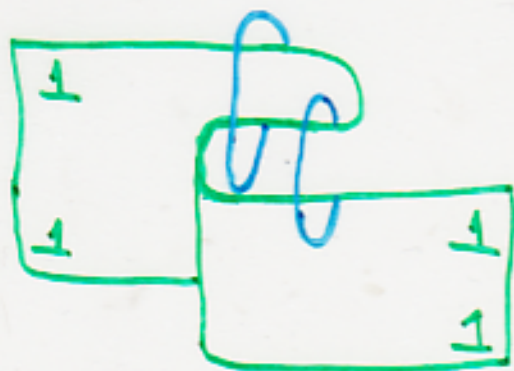
- ⇒ The above figure contains two types of nodes
- those which have an even number of edges and
 - those which have an odd number of edges.
- ⇒ If the "trace" is to enter a node and then leave it (with no overlaps allowed) then there must be an even number of edges emanating from that node.
- ⇒ If instead there are an odd number of edges emanating then either
- the trace will not begin at that node but, at some time, the trace will arrive at that node and never leave or
 - the trace will begin at that node but not end there

⇒ The drawing in above figure has four nodes from which an odd number of edges emanate. The trace must either begin or end at each one of these. That is clearly impossible.



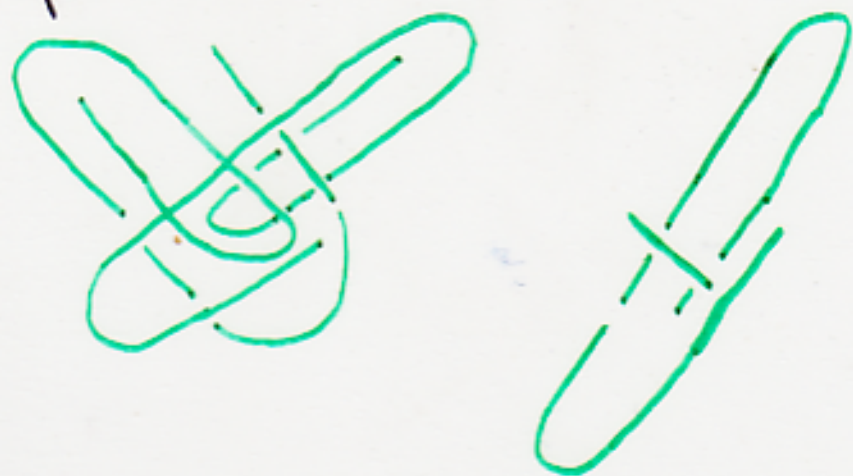
REAL LIFE PROBLEMS

① Examine the configuration of two paper clips and a dollar bill shown in Figure below. If the two ends of the dollar bill are pulled apart sharply, then the paper clips will spring free of the dollar bill and will be linked. Explain why they become linked.



⇒ You are pulling each paper clip b/w two lines of the other paper clip, thus linking them. The final jerk of the paper [PAGE 96]

tosses the front paper clip over the top of the dollar bill to join the other paper clip in back, thus freeing the linked clips.



② ~~Look~~ at ~~Figure~~ belows. Notice that ~~Volume~~ 1 is on the left and ~~volume~~ 2 is on

② Two volume of Gibbon's The Rise and Fall of the Roman Empire sit in order on the shelf. The cover material of these books is $\frac{1}{8}$ " inch thick. The 500 pages in each volume are $\frac{1}{2}$ " thick. A small worm bores a hole and crawls from the very first page of Volume 1 to the very last page of Volume 2. How far does the worm crawl?



Notice that Volume 1 is on the left and Volume 2 is on the right. Notice that the first page of Volume 1

is just left of center and the last page of Volume 2 is just to the

right of center. The worm only has to bore from the inside front cover of Volume 1 to the outside (b/w the two books) and then bore into the back cover of volume 2. The worm only bores through two book covers, and no pages. So, the worm travels a total of $\frac{1}{4}$ ".

③ Say that a bottle has a round or square, flat, bottom and it has straight sides. The bottle is partly full (about half) of liquid. It is tapered at the top (as bottles usually are) and has a screw-on cap. How can you accurately determine the volume of the bottle if you are equipped with only a ruler?



Round Bottom



Square Bottom

⇒ Use the ruler to measure the base. Calculate the area. ~~Calculate~~ Call that A . Now measure the height h of the water. Then volume of the water in the bottle is $V = A \cdot h$.

⇒ Now turn the bottle upside down. We see that the water fills up the (difficult to measure) neck, and that the part not filled by water is cylindrical. Now, measure the height h' of the air column.

inside the bottle and above the water.
The volume of that air column is

$$V' = A \cdot h'$$

⇒ The total volume inside the bottle is
then just $V = V + V'$.



- ④ A new car is equipped with three fuel saving devices. Device A, by itself, saves 25% on fuel; device B, by itself, saves 45% on fuel; and device C, by itself, saves 30% on fuel.

Now suppose that the three devices are used together, and that they act independently. Will the combination save $25 + 45 + 30 = 100$ percent on fuel? Probably not. What is the correct answer.

⇒ Device A results in using only .75 of the fuel that the machine would use without any devices. Device B applied on top of device A results in .55 of that fuel being used to achieve the same mileage and device C on top of those two results in .70 of that fuel being used to achieve the same mileage.

All told, equipped with devices A, B and C, we need only use $.75 \times .55 \times .70$

as much fuel to achieve the same mileage as the car without any fuel saving devices. That is, we need only use .28875 as much fuel. Thus, in total, we are saving .71125 or 71.125% on fuel consumption.

⑤ A martini is made by mixing K parts gin with 1 part vermouth. Gin is usually 40% alcohol while vermouth is 20% alcohol. A martini is said to be "dry" if it contains relatively little vermouth. For instance if $K=15$ then the martini is said to be dry. If instead $K=5$ then the martini is said to be "sweet".

Some people refuse to drink dry martinis, as they claim it makes them drunk very quickly. Others prefer dry martinis because they taste better. Shed some light on this discussion by calculating the amount of alcohol in a dry martini vs. a sweet martini as described in the last paragraph.

Sweet Martini: $K=5$

⇒ It is made up of 6 parts - 5 of them gin and 1 of them vermouth.

Total alcohol^{parts} contain in sweet martini
 $= 40\% \text{ of } 5 + 20\% \text{ of } 1 = 2 + 0.2 = 2.2 \text{ parts}$

Hence in sweet martini, we have [PAGE 100]

altogether 2.2 out of 6 parts alcohol. Thus, the percentage of alcohol in a "sweet martini" is $2.2/6$ or 36.67% alcohol.

Similarly, for dry martini, $K=16$

⇒ It is made up of 16 parts - 15 of them gin and 1 of them vermouth.

Total alcohol parts contain in dry martini
 $= 40\% \text{ of } 15 + 20\% \text{ of } 1 = 6 + 0.2 = 6.2 \text{ parts.}$

⇒ Hence in dry martini, we have altogether 6.2 out of 16 parts alcohol. Thus, the percentage of alcohol in a "dry martini" is $\frac{6.2}{16}$ or 38.75% alcohol.

⇒ Difference in alcohol content b/w the sweet and dry martini is about 2.09%

Statistics Problem :→

⇒ It is claimed that the average child has no time to go to school. For the child spends 8 hours per day, or one third of his/her time sleeping. Based on a 365 day year, that's 121.67 days sleeping.

Also the child spends three hours per day eating. That's a total of 45 days in the year spent eating.

Also the child spends 90 days taking summer vacation.

Also the child spends 21 days on Christmas and Easter holiday

Finally, the child has each Saturday and Sunday off. That's a total of 104 days
⇒ In short, we have (rounding to whole days) accounted for

$$122 + 45 + 90 + 21 + 104 = 382 \text{ days}$$

of the year taken up by ordinary child-like activities. This is already more than the 365 days that are known to comprise a year. We conclude that there is certainly no time for the child to attend the school. What is wrong with this reasoning?

⇒ During the 90 days that the child has summer vacation, the child is also eating and sleeping. Thus, those hours have been counted twice. Likewise, the child eats and sleeps on weekends.

② A certain small business shows the following data at the end of a certain fiscal year. There are 4 owners and 120 employees. Each owner receives \$100,000 per year in annual salary. Each employee is paid \$12,000 per year in annual wages. There is a total of \$240,000 profit, to be divided among the owners. Set up two different statistical models which could be used to generate a report on the finances of this company.

→ The most simple-minded report (and perhaps the most truthful) is this

① Each employee receives \$12,000—that is all.

② Each owner receives \$100,000 in salary and \$60,000 in profits for a total of \$160,000.

→ Each owner is to be remunerated at a rate that is about 13.333 times the rate at which each employee is remunerated. Moreover, each owner receives profits equal to 500% of the wages of a typical employee. These numbers makes for bad public relations, from the point of view of management. There are a great many ways to do book keeping.

⇒ Suppose that we say that \$160,000 of the profits will be used as salary bonuses for the owners. Thus, each owner will receive \$100,000 in regular salary, \$40,000 in salary bonus (for a total of \$140,000 salary for each owner) and \$20,000 in profits. Look how advantageously this may be reported.

$$\begin{aligned}\text{average salary} &= \frac{120 \cdot 12,000 + 4 \cdot 140,000}{120 + 4} \\ &= \$16,129.03\end{aligned}$$

And the profit per owner is now \$20,000.

⇒ The average salary is $\frac{4}{3}$ of what each ~~worker~~ ^{worker} makes - but that is reasonable because the owners will (and should) make more than the workers. And each owner ~~makes~~ receives \$20,000 in profit. But that makes a total of \$80,000 in profits; compare ^{that} with \$2 million dollars paid in salaries.

⇒ We see that a very small total profit is being made as compared to the amount of salaries paid.

⇒ Clearly, this is a company that is primarily interested in the welfare of its employees, and only just a little in the wealth of its owners

⇒ Imagine that a certain election has three candidates: A, B, C. Now we describe the three voting systems as they would apply to these candidates.

Method (i) is the simplest, and perhaps the most flawed: the candidate with the greatest number of votes wins. ~~the~~ (Plurality method).

In Method (ii), each voter ranks the candidates first, second, and third. The "average rank" of each candidate is calculated, and the candidate with the greatest average wins. (Borda method).

In Method (iii), each voter ranks the candidates first, second, and third. If no candidate gets a clear majority of firsts, then the candidate with the fewest firsts is eliminated. Each of that candidate's votes is given to the candidate whom each voter ranked second. (Hare method).

⇒ Imagine an election involving three candidates A, B, C. There are 33 voters, so that we may treat all voting methods at once, we suppose that every voter votes, and that every voter gives a linear ranking to 3 candidates.

Ranking	No. of Voters
ABC	10
ACB	4
BAC	2
BCA	7
CAB	3
CBA	7

Method (i) Fourteen voters prefer A, 9 voters prefer B and 10 voters prefer C. The candidates with the greatest number of votes is A. Thus, A wins the election by the plurality method.

Method (ii) (Borda method)

⇒ We assign a candidate 3 points for each voter who ranks him first, 2 points for each voter who ranks him second, and 1 point for each voter who ranks him third.

Total number of points for A

$$= (10 \times 3) + (4 \times 3) + (2 \times 2) + (3 \times 2) + (7 \times 1) + (7 \times 1) = 66$$

Total number of points for B

$$= (10 \times 2) + (4 \times 1) + (2 \times 3) + (7 \times 2) + (3 \times 1) + (7 \times 2) = 68$$

Total number of points for C

$$= (10 \times 1) + (4 \times 2) + (2 \times 1) + (7 \times 2) + (3 \times 3) + (7 \times 3) = 64$$

Plainly, A's ^{average} rank is $66/33$, B's average rank is $68/33$ and C's average rank is $64/33$. With this method, B wins.

Method (iii) Hare method

⇒ Plainly no candidate has a majority of first place votes. We eliminate B, because B has the fewest first place votes. Two of B's first place votes rank A second, so we give those two votes to A. Seven of B's first place votes rank C second, so we give those seven votes to C. Thus, A ends up with $14+2=16$ votes, and C ends up with $10+7=17$ votes. As a result C wins.